

NETWORKWALKS
CYBERSECURITY & ETHICAL HACKING INTERNSHIP

WEEK 2 PROJECT REPORT

Footprinting, Reconnaissance and Network Scanning

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Program: Cybersecurity & Ethical Hacking Internship

Week: 2

Date: 19 September 2026

1. Scope and Authorization

This report documents the Week 2 practical activities completed using the NetworkWalks project modules. The supplied authorization material defines the permitted scope as the networkwalks.com public website and its associated DNS records, together with the tester's own local area network. It also limits the work to reconnaissance, passive/light information gathering, and network host discovery. Exploitation, unauthorized access, brute-force activity, disruptive testing, and interaction with out-of-scope systems are excluded.

The practical work was therefore performed with an emphasis on evidence collection, passive OSINT, light reconnaissance, and a Ping Scan of the tester-controlled LAN. Where a module's example requested an external organization that was not within the authorized scope, the procedure was adapted to the authorized networkwalks.com domain rather than testing the unrelated organization.

2. Executive Summary

During Week 2, I completed practical exercises covering five project modules: footprinting with multiple Kali Linux tools, Google Hacking Database (GHDB) reconnaissance, Maltego-based email footprinting, theHarvester-based public-source enumeration, and Zenmap network scanning. The work demonstrated how publicly exposed information and local network discovery can be collected in a controlled and authorized manner.

The footprinting activity against networkwalks.com produced domain, DNS, HTTP-header, WAF, and DNS-record observations. GHDB work documented camera-related search dorks and mathematics PDF directory listings without opening live third-party camera endpoints. Maltego successfully returned sales@networkwalks.com and info@networkwalks.com through the configured search-engine transform, while a WHOIS transform returned abuse@godaddy.com as a registrar/abuse contact. theHarvester returned public-source information including info@networkwalks.com, multiple subdomains/hosts, and two visible IP addresses. Zenmap identified three live hosts on the tester-controlled 192.168.0.0/24 LAN and generated a topology view.

No exploitation, credential attacks, denial-of-service activity, privilege escalation, or unauthorized access was performed.

3. Objectives

- Perform footprinting and reconnaissance against the authorized networkwalks.com domain.
- Use GHDB to understand how search operators can surface publicly indexed information.
- Use Maltego to map a domain and identify related email entities.
- Use theHarvester to collect publicly available email and host/subdomain information.
- Use Zenmap to discover live hosts on the tester-controlled LAN.
- Record commands, observations, evidence, limitations, and security implications.

4. Tools and Technologies Used

Tool / Technology	Purpose
Kali Linux	Reconnaissance and footprinting environment.
WHOIS	Public domain registration and name-server information.
WhatWeb	Web-technology fingerprinting; execution was tested but produced no usable output in this environment.
Nslookup	DNS resolution of the target domain.
Curl	Inspection of HTTP response headers.
Wafw00f	Identification of Web Application Firewall technology.
DNSRecon	Enumeration of NS, MX, TXT/SPF and SRV records.
Exploit-DB GHDB	Collection of Google dorks for search-based reconnaissance.
Maltego Graph 4.13.0	Domain-to-email entity mapping using transforms.
theHarvester 4.10.1	Public-source email, host and subdomain enumeration.
Zenmap / Nmap 7.99.1	Ping Scan and topology visualization of the local LAN.
Windows CMD / ipconfig	Identification of local IPv4 address, subnet mask and gateway.

5. Activities Performed

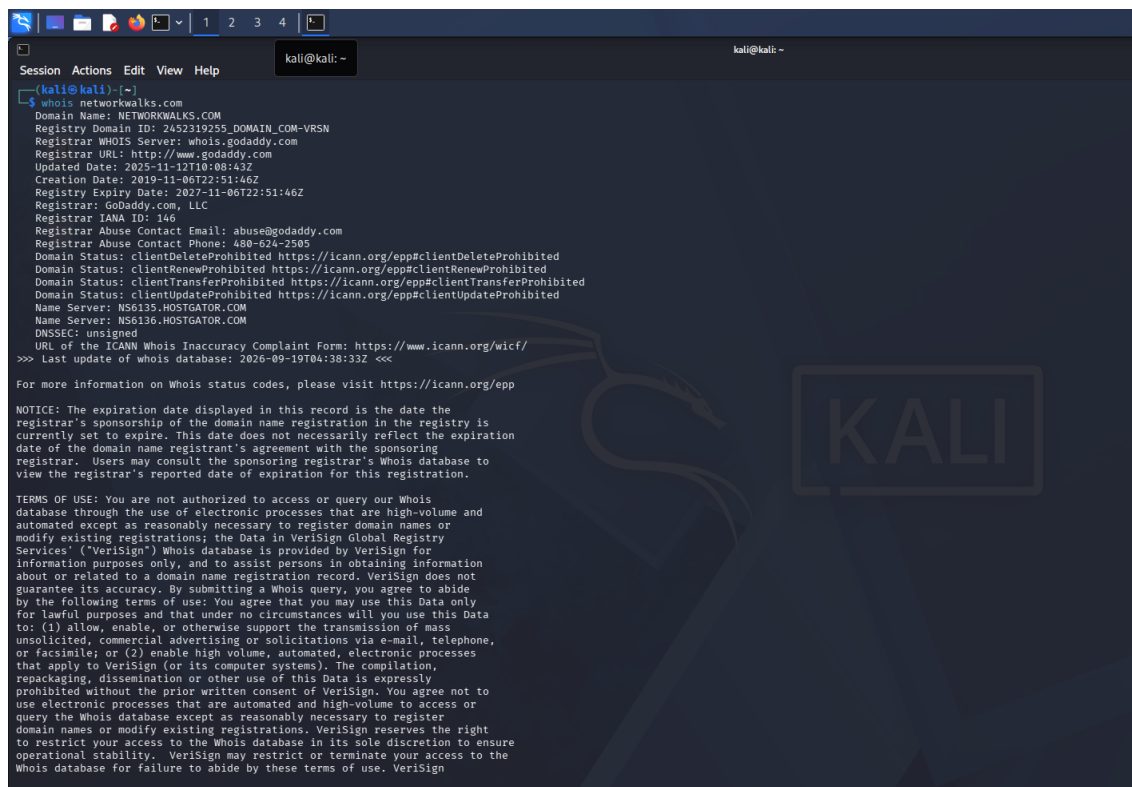
5.1 Project Module 1 — Footprinting with Multiple Kali Tools

The module requires six reconnaissance activities against networkwalks.com: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. The procedures and terminology follow the supplied NetworkWalks module.

5.1.1 WHOIS

whois networkwalks.com

The WHOIS output was captured to a text file. The result exposed public domain-registration information and name-server information. The observed name servers included ns6135.hostgator.com and ns6136.hostgator.com, providing an indication of the DNS/hosting infrastructure.

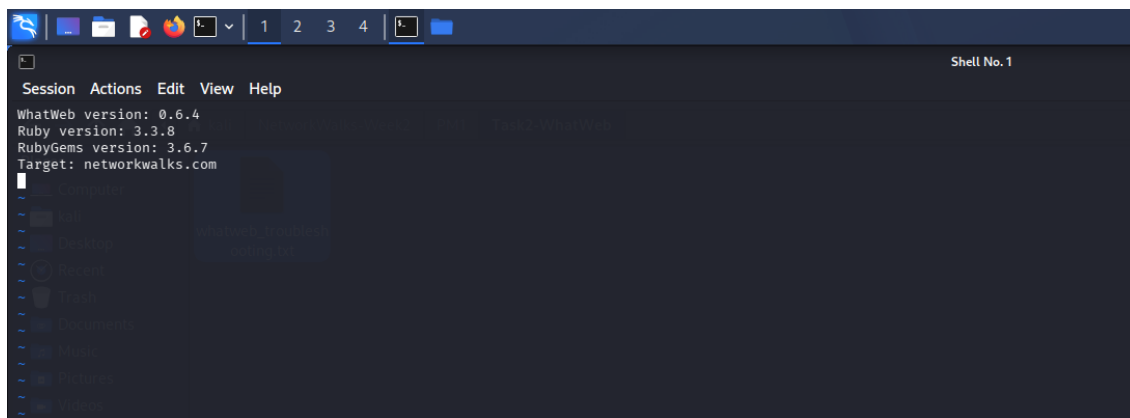


```
kali@kali: ~  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 245219255-DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-09-19T04:38:33Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry  
Services' ("VeriSign") Whois database is provided by VeriSign for  
information purposes only, and to assist persons in obtaining information  
about or related to a domain name registration record. VeriSign does not  
guarantee its accuracy. By submitting a Whois query, you agree to abide  
by the following terms of use: You agree that you may use this Data only  
for lawful purposes and that under no circumstances will you use this Data  
to: (1) allow, enable, or otherwise support the transmission of mass  
unsolicited, commercial advertising or solicitations via e-mail, telephone,  
or facsimile; or (2) enable high volume, automated, electronic processes  
that apply to VeriSign (or its computer systems). The compilation,  
repackaging, dissemination or other use of this Data is expressly  
prohibited without the prior written consent of VeriSign. You agree not to  
use electronic processes that are automated and high-volume to access or  
query the Whois database except as reasonably necessary to register  
domain names or modify existing registrations. VeriSign reserves the right  
to restrict your access to the Whois database in its sole discretion to ensure  
operational stability. VeriSign may restrict or terminate your access to the  
Whois database for failure to abide by these terms of use. VeriSign
```

5.1.2 WhatWeb

whatweb networkwalks.com

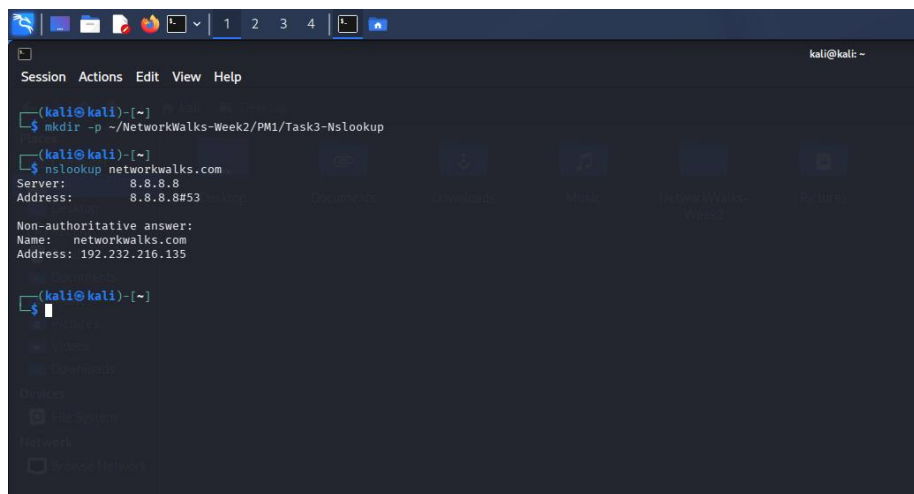
WhatWeb was executed, but the installed version did not return usable target output within the testing window. A timeout/debug check was also performed and recorded. Because no reliable output was obtained, no technology/version finding has been asserted from WhatWeb in this report.



5.1.3 Nslookup

```
nslookup networkwalks.com
```

Nslookup resolved networkwalks.com to 192.232.216.135. This provides the public IPv4 address associated with the target at the time of testing.



5.1.4 Curl

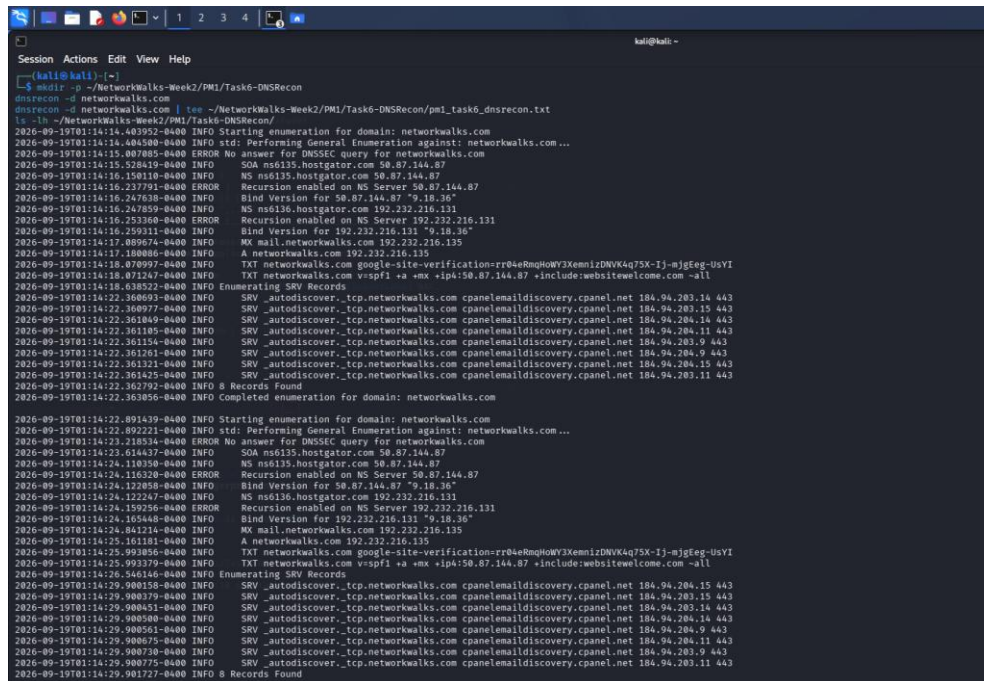
```
curl -I https://networkwalks.com
```

The HTTP header response returned HTTP/2 200. The response exposed Apache in the server header, an x-nginx-cache value indicating WordPress caching infrastructure, content-type text/html, and a WordPress REST API reference. These are information-disclosure observations rather than confirmed vulnerabilities.

5.1.6 DNSRecon

```
dnsrecon -d networkwalks.com
```

DNSRecon identified SOA/NS records, the mail host, the public A record, TXT records including SPF-related information, and cPanel-related SRV records. The observed SOA/name-server information included ns6135.hostgator.com (50.87.144.87) and ns6136.hostgator.com (192.232.216.131). The mail host was mail.networkwalks.com at 192.232.216.135. Eight records were reported by the tool in the captured output.



```
kali@kali: ~  
[kali@kali]~  
$ cd /p/.NetworkWalks-Week2/PM1/Task6-DNSRecon  
$ dnsrecon -d networkwalks.com  
dnsrecon -d networkwalks.com 1 2 3 4  
$ cd /p/.NetworkWalks-Week2/PM1/Task6-DNSRecon  
$ dnsrecon -d networkwalks.com  
2026-09-19T01:14:14.403952-0400 INFO Starting enumeration for domain: networkwalks.com  
2026-09-19T01:14:14.404500-0400 INFO std: Performing General Enumeration against: networkwalks.com...  
2026-09-19T01:14:15.007805-0400 ERROR No answer for DNSSEC query for networkwalks.com  
2026-09-19T01:14:15.528419-0400 INFO SOA ns6135.hostgator.com 50.87.144.87  
2026-09-19T01:14:16.150110-0400 INFO NS ns6135.hostgator.com 50.87.144.87  
2026-09-19T01:14:16.227701-0400 ERROR Recursion enabled on NS Server 50.87.144.87  
2026-09-19T01:14:16.247638-0400 INFO Bind Version for 50.87.144.87 "9.18.36"  
2026-09-19T01:14:16.247859-0400 INFO NS ns6136.hostgator.com 192.232.216.131  
2026-09-19T01:14:16.253300-0400 ERROR Recursion enabled on NS Server 192.232.216.131  
2026-09-19T01:14:16.259311-0400 INFO Bind Version for 192.232.216.131 "9.18.36"  
2026-09-19T01:14:17.089674-0400 INFO MX mail.networkwalks.com 192.232.216.135  
2026-09-19T01:14:17.100080-0400 INFO A networkwalks.com 192.232.216.135  
2026-09-19T01:14:18.070997-0400 INFO TXT networkwalks.com google-site-verification=rr04eRngHOWV3XennizDNVXKq75X-1j-mjgteg-UsVI  
2026-09-19T01:14:18.071247-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com -all  
2026-09-19T01:14:18.618522-0400 INFO Enumerating SRV Records  
2026-09-19T01:14:22.308603-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443  
2026-09-19T01:14:22.308977-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443  
2026-09-19T01:14:22.361049-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443  
2026-09-19T01:14:22.361195-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443  
2026-09-19T01:14:22.361134-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443  
2026-09-19T01:14:22.361261-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443  
2026-09-19T01:14:22.361321-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443  
2026-09-19T01:14:22.361425-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443  
2026-09-19T01:14:22.362792-0400 INFO 8 Records Found  
2026-09-19T01:14:22.363856-0400 INFO Completed enumeration for domain: networkwalks.com  
2026-09-19T01:14:22.891439-0400 INFO Starting enumeration for domain: networkwalks.com  
2026-09-19T01:14:22.892221-0400 INFO std: Performing General Enumeration against: networkwalks.com...  
2026-09-19T01:14:23.118314-0400 ERROR No answer for DNSSEC query for networkwalks.com  
2026-09-19T01:14:23.614437-0400 INFO SOA ns6135.hostgator.com 50.87.144.87  
2026-09-19T01:14:24.110358-0400 INFO NS ns6135.hostgator.com 50.87.144.87  
2026-09-19T01:14:24.116338-0400 ERROR Recursion enabled on NS Server 50.87.144.87  
2026-09-19T01:14:24.122858-0400 INFO Bind Version for 50.87.144.87 "9.18.36"  
2026-09-19T01:14:24.122247-0400 INFO NS ns6136.hostgator.com 192.232.216.131  
2026-09-19T01:14:24.159236-0400 ERROR Recursion enabled on NS Server 192.232.216.131  
2026-09-19T01:14:24.165448-0400 INFO Bind Version for 192.232.216.131 "9.18.36"  
2026-09-19T01:14:24.841214-0400 INFO MX mail.networkwalks.com 192.232.216.135  
2026-09-19T01:14:25.161181-0400 INFO A networkwalks.com 192.232.216.135  
2026-09-19T01:14:25.993856-0400 INFO TXT networkwalks.com google-site-verification=rr04eRngHOWV3XennizDNVXKq75X-1j-mjgteg-UsVI  
2026-09-19T01:14:25.993856-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com -all  
2026-09-19T01:14:26.546146-0400 INFO Enumerating SRV Records  
2026-09-19T01:14:29.900316-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443  
2026-09-19T01:14:29.900379-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443  
2026-09-19T01:14:29.900431-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443  
2026-09-19T01:14:29.900500-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443  
2026-09-19T01:14:29.900501-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443  
2026-09-19T01:14:29.900675-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443  
2026-09-19T01:14:29.900730-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443  
2026-09-19T01:14:29.900775-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443  
2026-09-19T01:14:29.901727-0400 INFO 8 Records Found
```

5.2 Project Module 2 — Footprinting and Reconnaissance with GHDB

The supplied GHDB module contains two tasks. Task 1 asks for live exposed security-camera links, while Task 2 asks for mathematics ebook PDF directory listings. Because the authorization scope does not permit interaction with third-party camera systems, the camera exercise was limited to documenting GHDB dorks and did not open or access live camera endpoints.

5.2.1 Task 1 — Camera-related GHDB dorks

The GHDB search for the term 'cam' produced camera-related entries. Ten representative entries were documented from the visible results:

No.	Documented GHDB dork
1	intitle:"Webcam" inurl:WebCam.htm
2	intitle:"Toshiba Network Camera"
3	intitle:"Index of /cam/"
4	intitle:"Index of /webcam/"
5	intitle:"Device(IP CAMERA)" "language" -com net
6	intitle:"NoVus IP camera" -com
7	intitle:"Network Camera" inurl:main.cgi
8	inurl:webcam site:skylinewebcams.com inurl:roma
9	intitle:"Login" intext:"cam"
10	intitle:"Real-time IP Camera Monitoring System" intext:"ActiveX Mode (For IE Browser)"

Additional visible GHDB camera entries were also recorded in the evidence file. No live camera endpoint was opened, logged into, downloaded from, or otherwise interacted with.

```

PS> kali@kali: /home/kali

Session Actions Edit View Help

>> PM2 TASK 1 - GHDB CAMERA SEARCH
>> GHDB Search Term:
>> cam
>> Search Result:
>> 100 GHDB entries found.
>> Evidence:
>> 01-GHDB-cam-search.png
>> Visible camera-related GHDB entries documented:
>> 1. Webcam / WebCam entry - Various Online Devices
>> 2. Toshiba Network Camera - Various Online Devices
>> 3. Index of /cam/ - Files Containing Juicy Info
>> 4. Index of /webcam/ - Files Containing Juicy Info
>> 5. Device (IP CAMERA) - Pages Containing Login Portals
>> 6. NoVus IP camera - Various Online Devices
>> 7. Network Camera - Various Online Devices
>> 8. Webcam site / ROMA - Various Online Devices
>> 9. Login / cam - Pages Containing Login Portals
>> 10. Real-time IP Camera Monitoring System - Various Online Devices
>> No live camera endpoints were opened or accessed.
>> @ | Out-File -Encoding UTF8 ~/NetworkWalks-Week2/PM2/Task1-Security-Cameras/pm2_task1_ghdb_camera_results.txt

(kali@kali)-[/home/kali]
PS> Get-Content ~/NetworkWalks-Week2/PM2/Task1-Security-Cameras/pm2_task1_ghdb_camera_results.txt
PM2 TASK 1 - GHDB CAMERA SEARCH
GHDB Search Term:
cam
Search Result:
100 GHDB entries found.
Evidence:
01-GHDB-cam-search.png
Visible camera-related GHDB entries documented:
1. Webcam / WebCam entry - Various Online Devices
2. Toshiba Network Camera - Various Online Devices
3. Index of /cam/ - Files Containing Juicy Info
4. Index of /webcam/ - Files Containing Juicy Info
5. Device (IP CAMERA) - Pages Containing Login Portals
6. NoVus IP camera - Various Online Devices
7. Network Camera - Various Online Devices
8. Webcam site / ROMA - Various Online Devices
9. Login / cam - Pages Containing Login Portals
10. Real-time IP Camera Monitoring System - Various Online Devices
No live camera endpoints were opened or accessed.
(kali@kali)-[/home/kali]
PS>

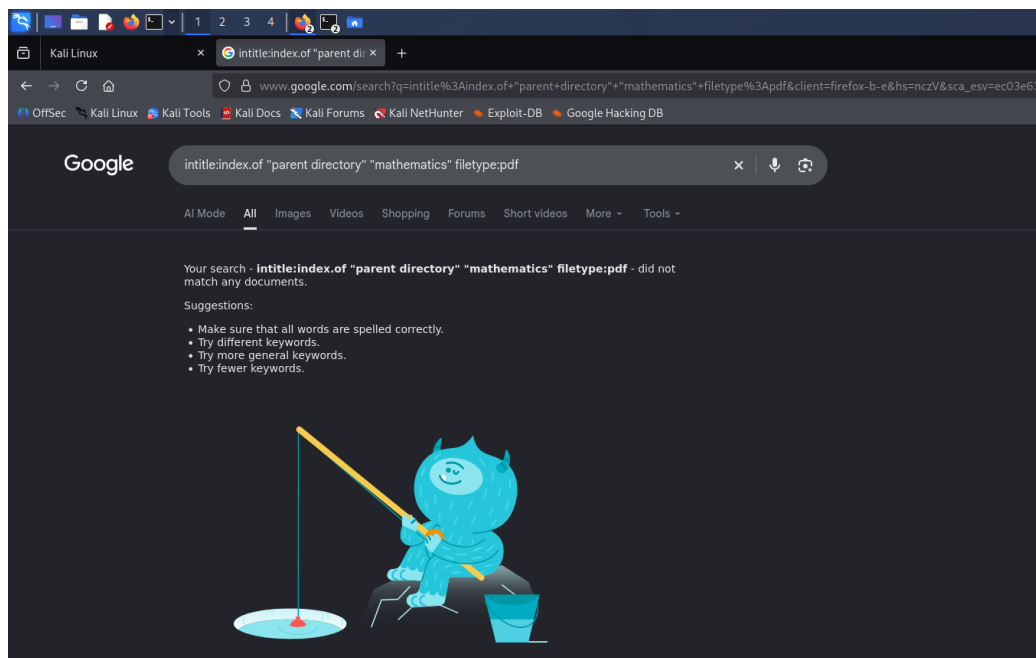
```

5.2.2 Task 2 — Mathematics PDF directory listings

Ten public directory/listing locations identified through search-based reconnaissance were documented. The purpose was to record the publicly indexed listing and the relevant search pattern; no copyrighted PDF collection was downloaded.

No.	Listing / Domain	Relevant search pattern
1	erwhon.superkuh.com/library/Math/	intitle:index.of "parent directory" mathematics pdf
2	sajairamcollege.ac.in/pdf/MATHEMATICS/	intitle:index.of "parent directory" mathematics pdf
3	theswissbay.ch/pdf/Books/Mathematics/	intitle:index.of "parent directory" mathematics pdf
4	isidore.co/misc/Physics papers and books/Mathematics/	intitle:index.of "parent directory" mathematics pdf
5	discrete.openmathbooks.org/pdfs/	intitle:index.of "parent directory" mathematics pdf
6	zaco.au/lib/math/hs/	intitle:index.of "parent directory" mathematics pdf
7	math.mcgill.ca/barr/	intitle:index.of mathematics pdf
8	unm.edu/~megrad/Math/	site:edu "index of" mathematics pdf
9	math.dartmouth.edu/~carlp/PDF/	site:edu "index of" mathematics pdf
10	docs.bartonccc.edu/syllabus/Master/MATH/	intitle:index.of mathematics pdf

Username/password fields were recorded as N/A where no credentials were involved.

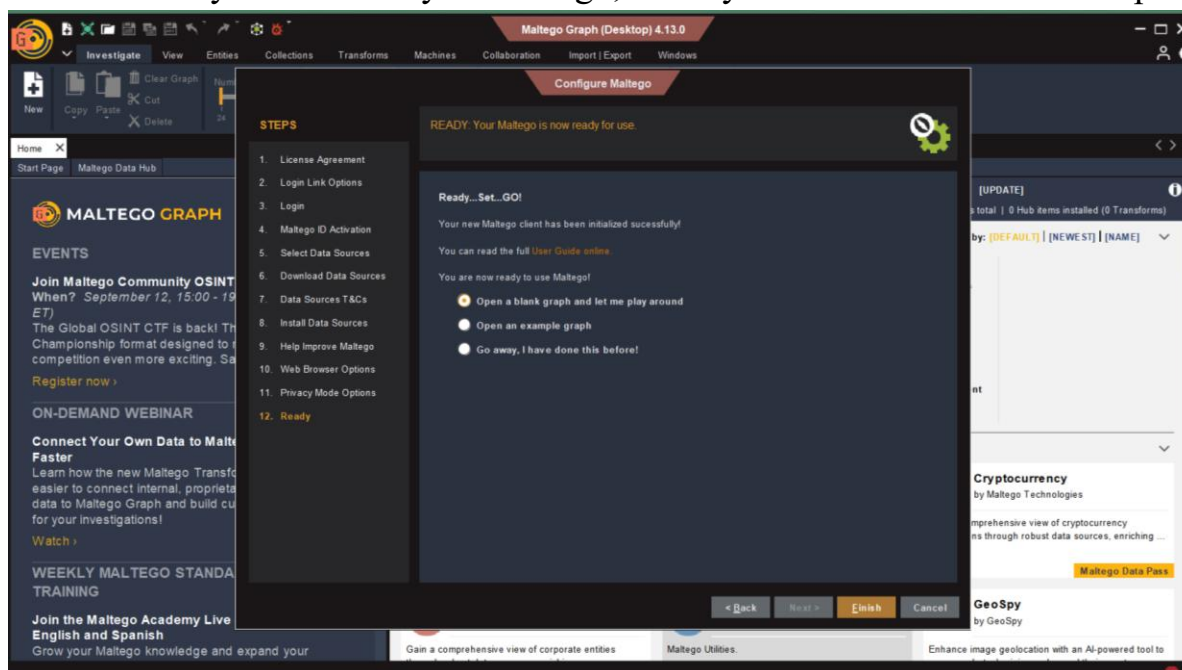


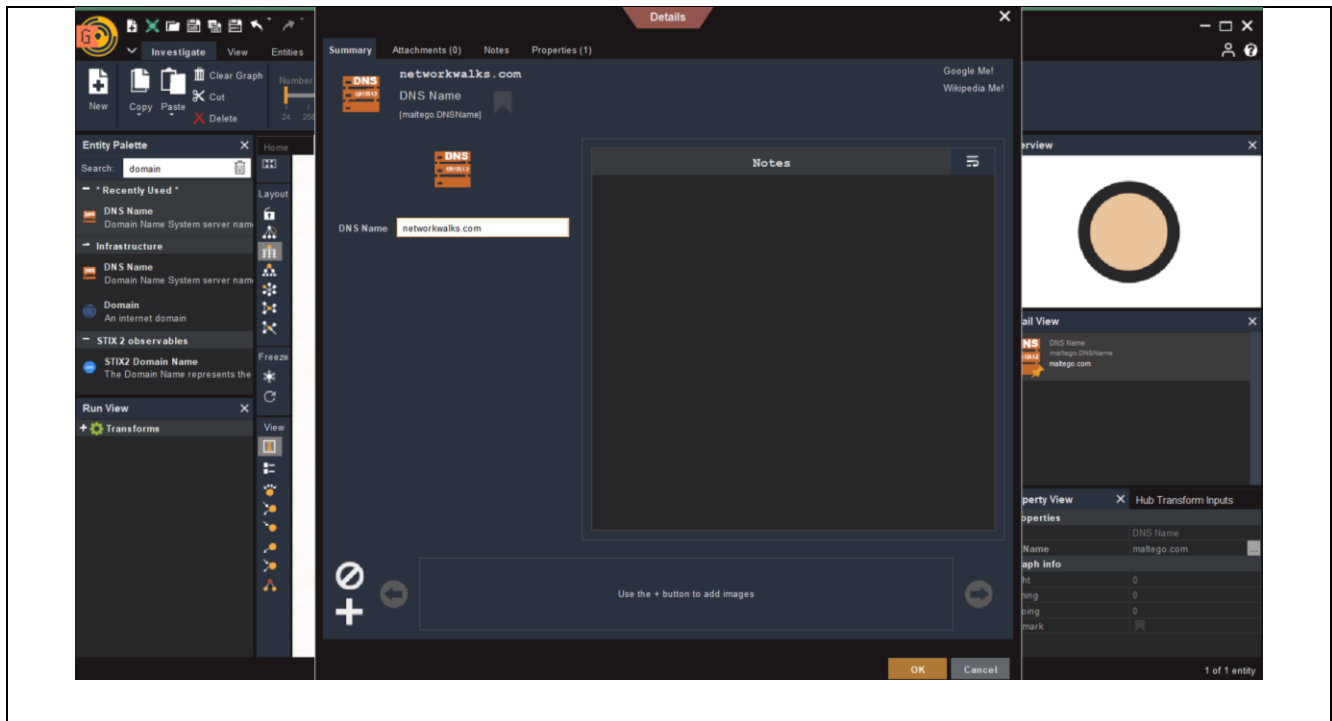
5.3 Project Module 3 — Footprinting with Maltego

The Maltego module requires installation on Windows and domain-to-email footprinting for networkwalks.com with permission. Maltego Graph 4.13.0 was installed and configured on Windows using the Maltego ID workflow.

5.3.1 Installation and configuration

1. Downloaded and installed Maltego Graph on Windows with the bundled Java runtime.
2. Completed Maltego ID/browser activation and initial profile configuration.
3. Initialized the desktop client and opened a blank graph.
4. Created a Domain entity and set it to networkwalks.com.
5. Configured the search-engine transform with a Google Programmable Search Engine ID and API key stored locally in Maltego; the key was not included in the report.





5.3.2 Email footprinting results

The correct Maltego entity was Domain: networkwalks.com. The following transforms were executed:

- [Utilities] To Email address [From whois info]
- [Utilities] To Email Addresses [PGP]
- [Utilities] To Emails @domain [Search Engine]

Transform	Observed result
WHOIS email transform	abuse@godaddy.com — registrar/abuse contact; not treated as a NetworkWalks organizational mailbox.
PGP email transform	No additional entities returned.
Search Engine email transform	sales@networkwalks.com and info@networkwalks.com.

The Search Engine transform completed successfully after configuration and returned two domain-related email entities. The Maltego graph was saved as PM3-NetworkWalks-Maltego-Email-Footprinting.mtgx.



```

Output - Transform Output
[9/19/26, 2:19 AM] WARNING Please input a valid 'engine ID' to use the Google API (from entity "networkwalks.com")
[9/19/26, 2:19 AM] INFO Transform [Utilities] To Emails $domain [Search Engine] completed in 2 s 692 ms (from entity "networkwalks.com")
[9/19/26, 2:19 AM] INFO Running transform [Utilities] To Email address [From whois info] on 1 entities (from entity "networkwalks.com")
[9/19/26, 2:19 AM] INFO ---[ Unified Credits ]--- : You have consumed 10 credits. You have 190 credits remaining. New period starts after October 1, 2026, 1
[9/19/26, 2:19 AM] INFO Transform [Utilities] To Email address [From whois info] completed in 1 s 927 ms with 2 entities returned (from entity "networkwalks.com")
[9/19/26, 2:20 AM] INFO Running transform [Utilities] To Email Addresses [PGP] on 1 entities (from entity "networkwalks.com")
[9/19/26, 2:20 AM] INFO ---[ Unified Credits ]--- : You have consumed 11 credits. You have 189 credits remaining. New period starts after October 1, 2026, 1
[9/19/26, 2:20 AM] INFO Transform [Utilities] To Email Addresses [PGP] completed in 2 s 72 ms (from entity "networkwalks.com")
[9/19/26, 2:48 AM] INFO Running transform [Utilities] To Emails $domain [Search Engine] on 1 entities (from entity "networkwalks.com")
[9/19/26, 2:48 AM] INFO ---[ Unified Credits ]--- : You have consumed 21 credits. You have 179 credits remaining. New period starts after October 1, 2026, 1
[9/19/26, 2:48 AM] WARNING Please input a valid 'engine ID' to use the Google API (from entity "networkwalks.com")
[9/19/26, 2:48 AM] INFO Transform [Utilities] To Emails $domain [Search Engine] completed in 2 s 339 ms (from entity "networkwalks.com")
[9/19/26, 3:19 AM] INFO Running transform [Utilities] To Emails $domain [Search Engine] on 1 entities (from entity "networkwalks.com")
[9/19/26, 3:19 AM] INFO ---[ Unified Credits ]--- : You have consumed 31 credits. You have 169 credits remaining. New period starts after October 1, 2026, 1
Running Transforms 0% 1 of 2 entities

```

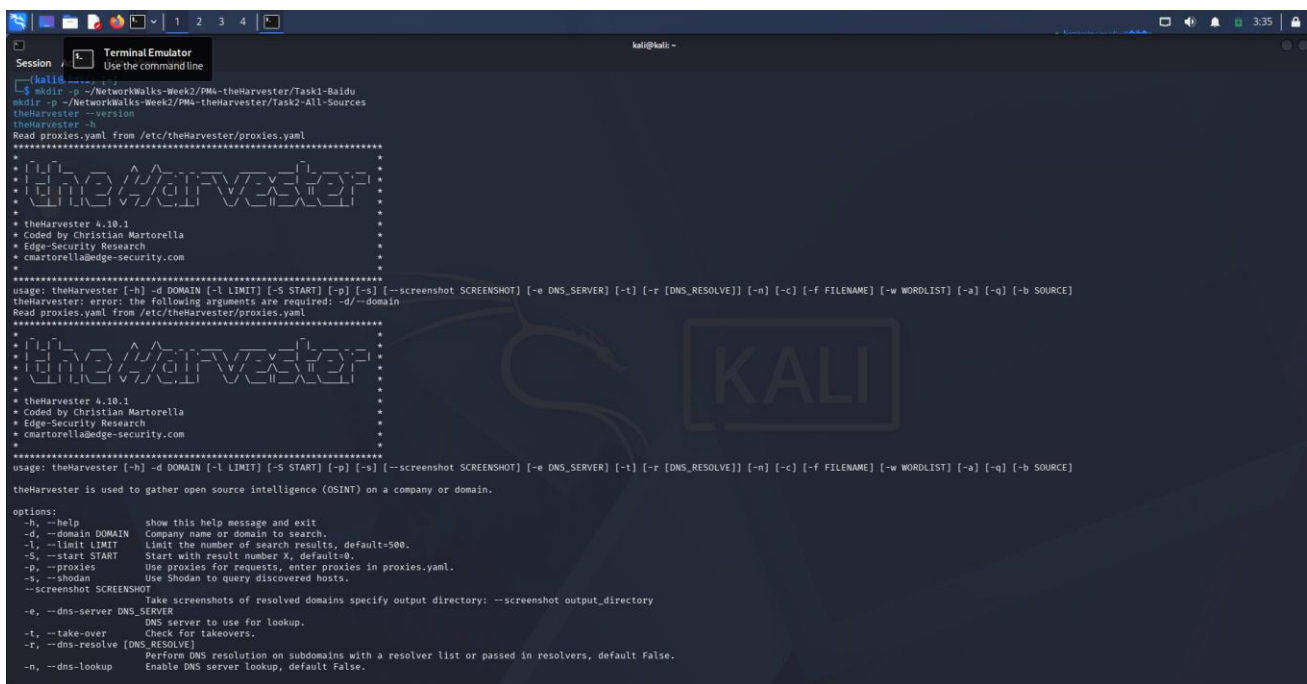
5.4 Project Module 4 — Footprinting and Reconnaissance with theHarvester

The supplied module uses microsoft.com as its example target. That target is outside the authorization scope used for this internship work. To keep the exercise within the permitted scope, the same theHarvester procedures were executed against networkwalks.com.

5.4.1 Task 1 — Baidu, limit 1000

```
theHarvester -d networkwalks.com -l 1000 -b baidu
```

The Baidu source returned no IPs, emails, people, or hosts in the captured run. The result is still valid evidence because public-source output can vary over time.



5.4.2 Task 2 — All sources, limit 50

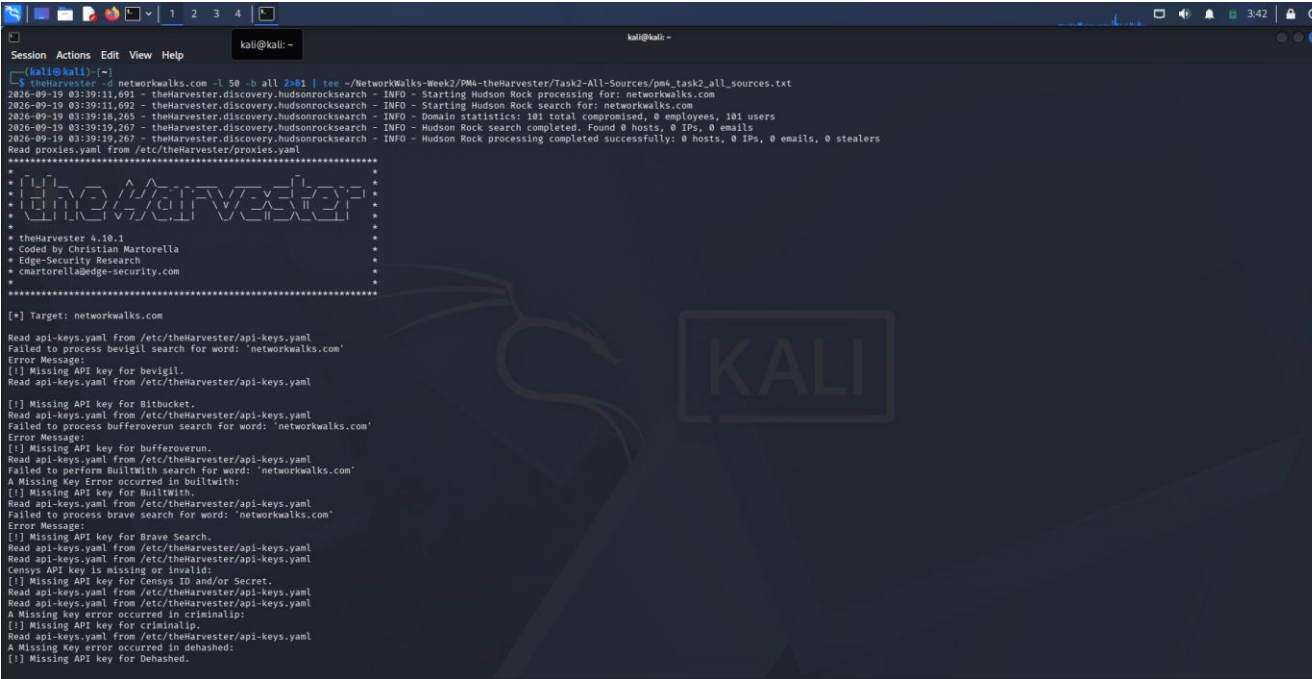
theHarvester -d networkwalks.com -l 50 -b all

The all-source run produced information from multiple public sources. Several sources were unavailable because they required API credentials, and those warnings were preserved in the evidence rather than being hidden.

Category	Observed result
Email	info@networkwalks.com
Visible IPs	172.67.198.228; 192.232.216.135
Hosts/subdomains	32 hosts reported by the tool
People	None reported
Interesting URLs	http://networkwalks.com/ ; https://networkwalks.com/
ASNs	AS13335, AS31898, AS46606
Sources	Multiple public sources including CRT.sh, Certspotter, DuckDuckGo, Common Crawl, OTX, RapidDNS, URLScan, Wayback Archive and others
API limitations	Several services reported missing/invalid API keys

Representative discovered hosts included autodiscover.networkwalks.com, cpanel.networkwalks.com, cpcalendars.networkwalks.com, cpcontacts.networkwalks.com, ftp.networkwalks.com, mail.networkwalks.com,

webdisk.networkwalks.com and webmail.networkwalks.com. The full tool output remains the authoritative evidence for the complete 32-host result.



5.5 Project Module 5 — Network Scanning with Zenmap

The NetworkWalks Zenmap module requires local IP/subnet identification, a Ping Scan, live-host counting, IP/MAC collection, and topology export. The supplied module explicitly notes that the example 10.0.0.0/24 network may differ from the tester's actual LAN.

5.5.1 Local network configuration

ipconfig

Parameter	Observed value
Active adapter	Wireless LAN adapter Wi-Fi
IPv4 address	192.168.0.184
Subnet mask	255.255.255.0
LAN subnet	192.168.0.0/24
Default gateway	192.168.0.1

The separate Ethernet 2 adapter showed 192.168.56.1 and was not selected for the LAN scan. The active Wi-Fi network was used.

```
C:\WINDOWS\system32\cmd. x + v
Default Gateway . . . . . : 192.168.0.1
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\dayus>ipconfig
Windows IP Configuration

Ethernet adapter Ethernet 2:
Connection-specific DNS Suffix . :
Link-Local IPv6 Address . . . . . : fe80::2e13:87b6:5130:7d7e%4
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Link-Local IPv6 Address . . . . . : fe80::7288:a7d7:e793:ae8b%3
IPv4 Address. . . . . : 192.168.0.184
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.0.1

Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\dayus>
```

Figure 1. Windows ipconfig output showing the active Wi-Fi configuration.

5.5.2 Zenmap Ping Scan

nmap -sn 192.168.0.0/24

Zenmap was configured with Target 192.168.0.0/24 and Profile 'Ping scan'. The scan completed successfully, scanning 256 addresses in approximately 3.34 seconds and identifying three live hosts.

Host	Status	MAC address	Vendor / note
192.168.0.1	Up	98:03:8E:0E:59:79	TP-Link Systems
192.168.0.193	Up	2E:22:C5:96:6D:65	Unknown
192.168.0.184	Up	Not displayed in the Ping Scan output	Tester's Windows host

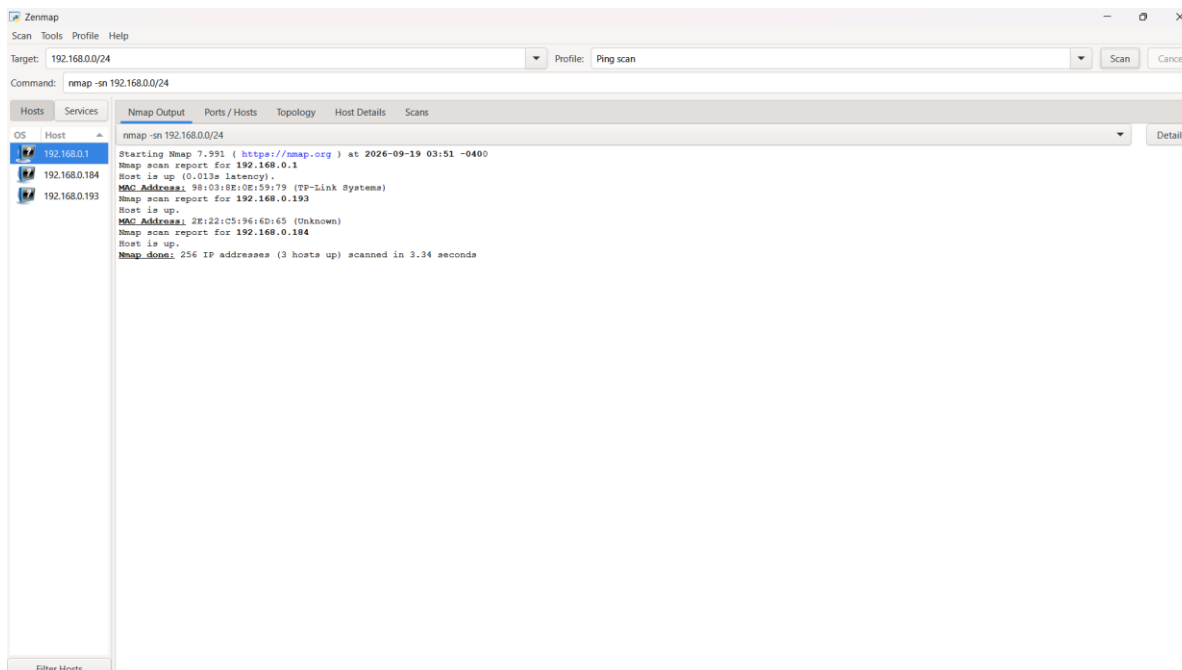


Figure 2. Zenmap Ping Scan showing three live hosts.

5.5.3 Topology

The Zenmap Topology view displayed the three discovered LAN hosts together with the local Zenmap/localhost node. The localhost node is not an additional scanned LAN host. The topology was saved as required by the module.

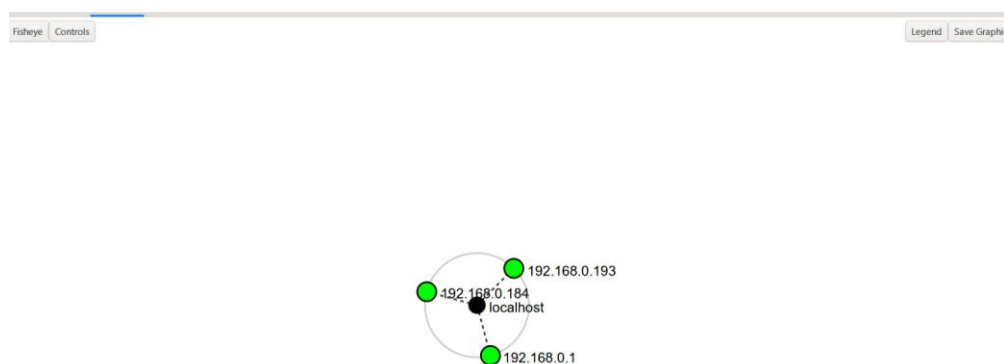


Figure 3. Zenmap Topology view for the 192.168.0.0/24 LAN.

The topology PDF is maintained with the project evidence as PM5-Zenmap-Network-Topology.pdf.

6. Risk Analysis / Impact

The observations below are information-security findings derived from reconnaissance and host discovery. They are not confirmed vulnerabilities. The risk descriptions are deliberately limited to what the collected evidence supports.

#	Observation	Evidence	Potential impact	Risk
1	Public DNS infrastructure is discoverable	WHOIS/DNSRecon exposed NS, MX, TXT/SPF and SRV information.	Supports external infrastructure mapping and service enumeration.	Medium
2	Web-service technical information is exposed	Curl returned HTTP/2 200, Apache, WordPress-related cache information and a REST API reference.	May assist technology fingerprinting and further authorized review.	Low
3	WAF technology is identifiable	Wafw00f identified ModSecurity (SpiderLabs).	Reveals part of the defensive architecture to an external observer.	Low
4	Multiple public hostnames are discoverable	theHarvester reported 32 hosts/subdomains.	Increases the amount of publicly visible infrastructure that should be reviewed and maintained.	Medium
5	Domain-related mailboxes are publicly discoverable	Maltego and theHarvester returned info@networkwalks.com and sales@networkwalks.com.	Publicly known mailboxes can attract unsolicited messages and targeted phishing attempts.	Low
6	Multiple live devices are visible on the tester LAN	Zenmap identified 3 live hosts on 192.168.0.0/24.	Unexpected devices on a local network should be identified and authorized.	Medium

No exploitation or vulnerability validation was performed. In particular, the GHDB camera exercise did not access live camera endpoints, and the theHarvester Microsoft example was not executed because Microsoft was outside the authorized scope.

7. Recommendations

6. Review public DNS records periodically and remove obsolete records or services.
7. Review publicly visible HTTP headers and application metadata for unnecessary technical disclosure.
8. Keep the WAF enabled and continuously review its configuration, logging and alerting.
9. Maintain an authoritative inventory of public subdomains and hosting endpoints and retire unused entries.
10. Use appropriate email security controls such as SPF, DKIM and DMARC and monitor public-facing mailboxes for abuse.
11. Perform regular authorized local-network discovery to identify unexpected devices.
12. Repeat passive reconnaissance periodically because search-engine indexes and public-source datasets change over time.
13. Keep reconnaissance evidence, API credentials and client information confidential and separate from public repositories.
14. Use written authorization and explicit scope boundaries for all future security testing.

8. Limitations and Evidence Notes

- WhatWeb did not produce usable output in the captured environment; no WhatWeb technology finding was invented or substituted.
- GHDB Task 1 was documented through dork discovery only; no live third-party camera endpoint was accessed.
- The mathematics PDF task documented public listings without downloading copyrighted PDF collections.
- Maltego search-engine transforms initially required configuration. After the Google Search Engine ID and API key were configured, the transform returned two domain email entities.
- theHarvester public-source results varied by source and several providers required API credentials. These warnings remain part of the evidence.
- The theHarvester module's example target microsoft.com was replaced with networkwalks.com to remain within the authorized scope.
- Zenmap was limited to the tester-controlled 192.168.0.0/24 LAN.
- The report records observations from the practical exercises and does not claim that any observation is a confirmed exploitable vulnerability.

9. Conclusion

During Week 2 of the NetworkWalks Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance, OSINT and local network scanning. The exercises demonstrated how different tools reveal different layers of an attack surface: WHOIS and DNSRecon exposed registration and DNS information; Nslookup and Curl exposed network and HTTP information; Wafw00f identified the visible WAF technology; GHDB demonstrated search-based discovery; Maltego mapped a domain to email entities; theHarvester aggregated public-source host and email information; and Zenmap identified active hosts on the tester-controlled LAN and produced a topology.

The work also reinforced the importance of authorization and scope control. Where a practical example requested an out-of-scope target or direct interaction with third-party systems, the procedure was adapted or limited so that no unauthorized access or interaction occurred. The resulting evidence provides a reproducible record of the authorized reconnaissance and scanning activities completed during Week 2.

10. Evidence Index

Module	Evidence / file	Purpose
PM1	PM1-Task1-WHOIS	WHOIS command and output
PM1	PM1-Task2-WhatWeb	WhatWeb execution/troubleshooting
PM1	PM1-Task3-Nslookup	DNS resolution
PM1	PM1-Task4-Curl	HTTP response headers
PM1	PM1-Task5-Wafw00f	WAF identification
PM1	PM1-Task6-DNSRecon	DNS enumeration
PM2	PM2-Task1-GHDB-Camera-Results	Camera-related GHDB dorks; no endpoint access
PM2	PM2-Task2-Math-PDF-Results	Ten mathematics PDF directory/listing results
PM3	PM3-Maltego-01-Installation.png	Installation/configuration evidence
PM3	PM3-Maltego-02-Domain-Entity.png	Domain entity evidence
PM3	PM3-Maltego-03-Email-Transforms.png	Email transform evidence
PM3	PM3-Maltego-04-WHOIS-Email.png	WHOIS-derived email evidence
PM3	PM3-Maltego-05-Google-Email-Results.png	Search-engine email results
PM3	PM3-NetworkWalks-Maltego-Email-Footprinting.mtgx	Saved Maltego graph
PM4	PM4-Task1-Baidu-Results.png / .txt	Baidu run
PM4	PM4-Task2-All-Sources-Results.png / .txt	All-source theHarvester run

PM5	PM5-01-ipconfig.png	Local IP/subnet evidence
PM5	PM5-02-ping-scan.png	Zenmap Ping Scan
PM5	PM5-03-topology.png	Zenmap topology
PM5	PM5-Zenmap-Network-Topology.pdf	Required topology PDF

All screenshots and text outputs should be retained with the corresponding module folders. Do not include private API keys or credentials in the final submission.

Appendix A — Key Commands

```
whois networkwalks.com
```

```
whatweb networkwalks.com
```

```
nslookup networkwalks.com
```

```
curl -I https://networkwalks.com
```

```
wafw00f https://networkwalks.com
```

```
dnsrecon -d networkwalks.com
```

```
theHarvester -d networkwalks.com -l 1000 -b baidu
```

```
theHarvester -d networkwalks.com -l 50 -b all
```

```
nmap -sn 192.168.0.0/24
```

Appendix B — PM5 Lab Answers

Question	Answer
Task 2: Local IP and LAN subnet	IPv4 192.168.0.184; subnet mask 255.255.255.0; subnet 192.168.0.0/24; gateway 192.168.0.1.
Task 4: How many hosts are live?	3 live hosts.
Task 5: IP addresses of live hosts	192.168.0.1; 192.168.0.193; 192.168.0.184.
Task 6: MAC addresses	98:03:8E:0E:59:79; 2E:22:C5:96:6D:65; the local Windows MAC was not shown in the Ping Scan output.
Task 7: Topology	Zenmap Topology view created and saved as a topology PDF.